



FILTERED REFERENCE AND FILTERED ERROR LMS ALGORITHMS FOR ADAPTIVE FEEDFORWARD CONTROL

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A unified and consistent formulation is developed for both filtered reference and filtered error forms of the instantaneous steepest descent, or LMS, algorithm when used to adapt FIR feedforward controllers. Both algorithms minimise the mean-square value of the same, output, error function. The two algorithms are first formulated for single-input single-output linear plants. It is argued that since the behaviour of the two algorithms is equivalent in the case of slow adaptation, the conditions on the accuracy of the plant model for stability should also be the same in both cases. This is expressed as a bound on the unstructured multiplicative uncertainty of the plant. Filtered reference and filtered error algorithms are also derived for multiple-input multiple-output (MIMO) linear systems, although the filtered reference algorithm is found not to have a simple block diagram interpretation. In the MIMO case, the filtered error form of the algorithm can have considerable computational advantages over the filtered reference form. Finally the two algorithms are extended to the case of non-linear plants and/or controllers which are modelled as feedforward neural networks. In the non-linear case the two formulations of the LMS algorithm reduce to two forms of the widely used backpropagation algorithm.

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1. INTRODUCTION

Adaptive feedforward techniques have been used widely in the active control of sound and vibration [1, 2], in which the control problem is generally one of disturbance rejection. They have also been used for the position control of flexible systems, which is a command-following servo-control task. The background and application of these methods in other application areas has recently been comprehensively described by Widrow and Walach [3], who term the techniques 'adaptive inverse control'. The object of this paper is to compare the filtered reference and filtered error forms of the instantaneous gradient descent, or LMS, algorithm when used for adaptive FIR, finite impulse response, feedforward controllers. Three cases are considered: the single and multichannel algorithms for linear systems and a general algorithm for non-linear systems. In each case the instantaneous steepest-descent algorithm can be formulated either in filtered reference form or in filtered error form, both of which minimise the mean-square output error. A wider range of algorithms has been reviewed by Sethares *et al.* [4], mostly for the adaptation of single channel IIR filters in electrical noise cancellation applications, in which the stabilising effects of either filtering the regressor or error have been compared. The present study concentrates on the case of FIR filters adapted using the LMS algorithm in which case the only need for filtering of the reference or error signals is to compensate for the response of the physical plant which links the output of the filters and

the observed error signals. The filtered error form of the LMS algorithm described here is somewhat different to the filtered- ε algorithm used by Widrow and Walach [3], and the reasons for this difference are examined. The connection between the filtered reference and filtered error algorithms is clearest for the single channel linear case, where they have simple block diagram interpretations. The filtered reference algorithm involves passing the reference signal through a model of the plant's impulse response and the filtered error algorithm involves passing the error signal through the impulse response of the plant model which has been reversed in time.

The algorithms can be extended to the multichannel case using a formulation involving multiple summations. These derivations lead to simple forms for the two algorithms, which can be readily implemented and have been found to work well in practice. The multichannel filtered reference algorithm does not appear to have a simple interpretation in terms of matrix block diagrams, whereas the multichannel filtered error algorithm does. For systems with multiple command inputs, the filtered error algorithm can also be considerably more efficient to implement than the multichannel filtered reference method.

When applied to non-linear systems, such as neural networks, the filtered reference algorithm can be seen to be a special form of real-time backpropagation (RTBP), and the filtered error algorithm as a special form of backpropagation through time (BPTT). The generation of the filtered error signal can then be interpreted as using the reciprocal version of the system under control. An attempt has been made to keep the notation in this paper consistent with the book by Widrow and Walach [3], in the hope of more clearly elucidating the formulation of these two complementary algorithms in the cases which they cover.

2. LINEAR SINGLE-INPUT SINGLE-OUTPUT PLANT

A simplified block diagram of the problem is shown in Fig. 1, in which a sampled reference, or command input, i_k , drives the plant $P(z)$ via a controller $\hat{C}(z)$. The objective is to minimise the overall system error, ε_k , which is the difference between the plant output and the desired signal d_k . Assuming the plant is represented by an FIR filter with coefficients $p_0 \cdots p_J$, where J may be arbitrarily large, and has an input signal u_k , then the overall error is equal to

$$\varepsilon_k = d_k - \sum_{j=0}^J p_j u_{k-j}. \quad (1)$$

If the controller is also an FIR filter with coefficients $\hat{c}_0 \cdots \hat{c}_N$ then the input to the plant is

$$u_k = \sum_{n=0}^N \hat{c}_n i_{k-n}, \quad (2)$$

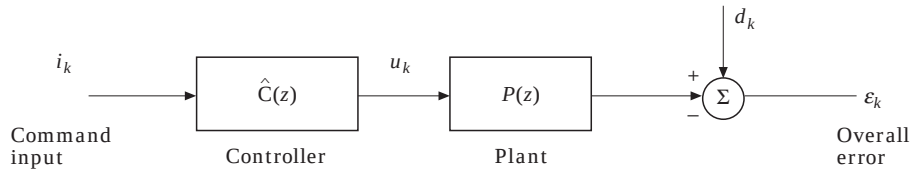


Figure 1. Block diagram of linear SISO control system.

so that the overall error may be expressed as the double summation

$$\varepsilon_k = d_k - \sum_{n=0}^N \sum_{j=0}^J \hat{c}_n p_j i_{k-n-j}. \quad (3)$$

The derivative of the time-averaged squared output error with respect to the n th controller coefficient is thus

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \varepsilon_k^2}{\partial \hat{c}_n} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \varepsilon_k \frac{\partial \varepsilon_k}{\partial \hat{c}_n} = - \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \varepsilon_k \sum_{j=0}^J p_j i_{k-n-j}, \quad (4)$$

which is used in two different ways for the filtered reference and filtered error adaptive algorithms. In all cases considered here, the controller coefficients are assumed to change only slowly with respect to the time constants of the plant dynamics, i.e. are quasi-static, and their time variation can thus be ignored in equation (4).

2.1. FILTERED REFERENCE ALGORITHM

The filtered reference (or filtered- x) LMS algorithm [5, 6, 7] can be obtained directly from equation (4) by defining

$$x'_k = \sum_{j=0}^J \hat{p}_j i_{k-j} \quad (5)$$

to be the reference signal filtered by a model which is an estimate of the true plant impulse response.

Updating the n th controller coefficient using an instantaneous estimate of equation (4) with this estimate of the filtered reference signal gives the practical filtered- x algorithm

$$\hat{\mathbf{c}}_{k+1} = \hat{\mathbf{c}}_k + \alpha \varepsilon_k \mathbf{x}'_k \quad (6)$$

where $\hat{\mathbf{c}}_k = [\hat{c}_{0,k} \ \hat{c}_{1,k} \ \cdots \ \hat{c}_{N,k}]^T$ is the vector of N controller coefficients at the k th time sample and

$$\mathbf{x}'_k = [x'_k \ x'_{k-1} \ \cdots \ x'_{k-N}]^T. \quad (7)$$

2.2. FILTERED ERROR ALGORITHM

Returning to equation (4), the derivative is now expressed another way by defining the dummy time variable

$$k' = k - j \quad \text{so that} \quad k = k' + j. \quad (8, 9)$$

Thus equation (4) can be written as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \varepsilon_k^2}{\partial \hat{c}_n} = - \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{j=0}^J \sum_{k'=k'+j=-K}^K i_{k'-n} \hat{p}_j \varepsilon_{k'+j}. \quad (10)$$

The error signal filtered *backwards* in time using the plant model, is now defined as

$$\delta_{k'} = \sum_{j=0}^J \hat{p}_j \varepsilon_{k'+j}. \quad (11)$$

Because j is always finite, the summation, from $-K$ to K as $K \rightarrow \infty$ on the right-hand side of equation (10), will be the same for $k' = -\infty$ to ∞ as for $k' + j = -\infty$ to ∞ . The time-averaged derivative in equation (10) can thus be expressed as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial e_k^2}{\partial \hat{c}_n} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \delta_k i_{k-n}. \quad (12)$$

Provided all the terms in equation (12) are used at different time steps to update the controller coefficients, and the adaptation rate is slow compared with J , the average adaptation behaviour will be similar to that when using equation (4), as noted by Wan [8]. The problem with using equation (12) in a real-time algorithm is that an instantaneous estimate of equation (12) cannot be implemented in a causal system. This problem can be overcome by delaying both δ_k and i_k by J samples.

The final filtered error adaptation algorithm can thus be written as

$$\hat{\mathbf{c}}_{k+1} = \hat{\mathbf{c}}_k + \alpha \delta_{k-J} \mathbf{i}_{k-J} \quad (13)$$

where

$$\mathbf{i}_{k-J} = [i_{k-J}, i_{k-J-1} \cdots i_{k-J-N}]^T \quad (14)$$

and the delayed filtered error signal can be written as

$$\delta_{k-J} = \sum_{j=0}^J \hat{p}_j e_{k+j-J} = \sum_{j'=0}^J \hat{p}_{J-j'} e_{k-j'}, \quad (15)$$

where the final form is obtained by letting $j' = J - j$, and emphasises that the error signal is now causally filtered using a time-reversed version of the model of the plant's impulse response. It should be noted that the filtered- ε algorithm discussed by Widrow and Walach [3] uses a delayed model of the *inverse* of the plant response to filter the error, rather than the delayed model of the *time-reversed* plant used above. The filtered- ε algorithm of Widrow and Walach minimises the mean square value of a *filtered* error signal, obtained by passing the output error through the inverse of the plant response, whereas the algorithm outlined above minimises the mean square value of the output error itself, and is thus more consistent with the widely used filtered reference LMS algorithm.

Block diagrams for the SISO filtered reference and filtered error algorithms are shown in Fig. 2, in which the forward plant model is denoted

$$\hat{P}(z) = \sum_{j=0}^J \hat{p}_j z^{-j} \quad (16)$$

and the delayed reverse plant model is denoted

$$\hat{P}_J(z^{-1}) = \sum_{j=0}^J \hat{p}_j z^{j-J} = \sum_{j'=0}^J \hat{p}_{J-j'} z^{-j'}, \quad (17)$$

where the final form of equation (17) has been obtained by again letting $j' = J - j$. Although the derivation above only shows that the filtered reference and filtered error algorithms are equivalent in the steady state, simulations by Wan [9] also suggests that they can have a similar transient performance. The speed of adaptation of the filtered error algorithm may, however, be reduced if the length of the plant model is too long, since a delay of this length is effectively introduced into the adaptation loop by the need to achieve

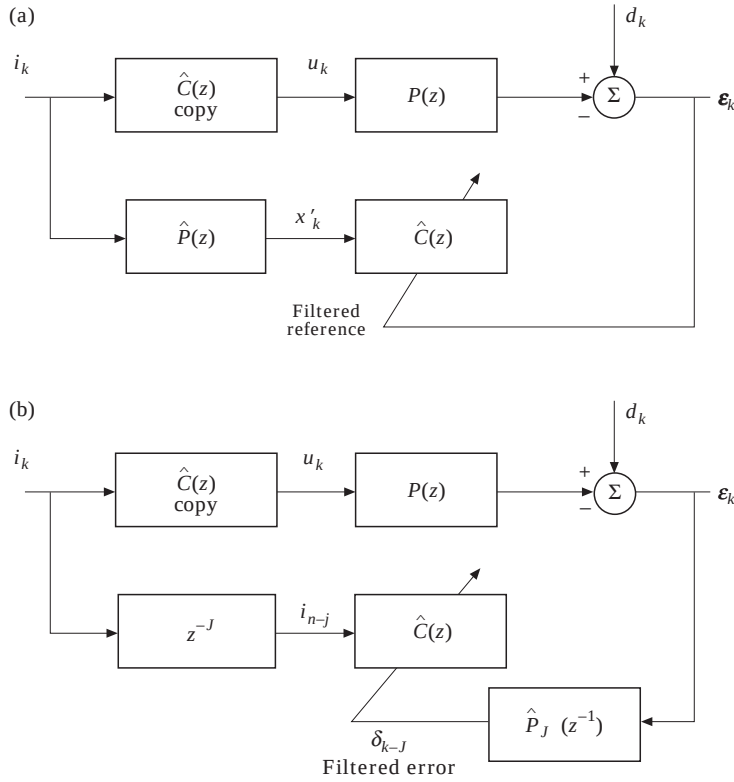


Figure 2. Block diagram of SISO (a) filtered reference and (b) filtered error controller adaptation algorithms.

a causal filtering of the error signal. Since the two algorithms are equivalent in the steady state, then for slow adaptation the conditions on the plant model for the algorithm to be stable will also be similar for the two cases. It was shown by Ren and Kumar [10] that a sufficient condition for the stability of the filtered-reference LMS algorithm is that the ratio of the transfer responses of the plant model and plant is strictly positive real (SPR), so the condition imposed on the frequency responses of the plant model and plant for stable adaptation is that

$$\operatorname{Re} \left[\frac{\hat{P}(e^{j\omega T})}{P(e^{j\omega T})} \right] > 0 \quad \text{for all } \omega T. \quad (18)$$

Equation (18) implies that the phase response of $\hat{P}(e^{j\omega T})$ must be within $\pm 90^\circ$ of the phase response of $P(e^{j\omega T})$ at all frequencies. The 90° phase condition has been used at individual frequencies to evaluate the stability of systems with tonal command signals for some time [5]. The condition imposed by equation (18) may also be cast in terms of a bound on the unstructured multiplicative uncertainty in the plant (see, for example, [11]). If the plant model is an accurate estimate of the nominal response of the plant, so that $\hat{P}(z) = P_0(z)$, and frequency response of the physical plant can be described by

$$P(e^{j\omega T}) = P_0(e^{j\omega T})[1 + \Delta p(e^{j\omega T})], \quad (19)$$

and the magnitude of $\Delta_p(e^{j\omega T})$ can be bounded at each ωT such that

$$|\Delta_p(e^{j\omega T})| \leq B(e^{j\omega T}) \quad (20)$$

where B is a real, frequency dependant number, but nothing is known about the phase of $\Delta_p(e^{j\omega T})$, then the uncertainty in the plant response is said to be unstructured. Assuming the uncertainty in the plant is described using equation (19), the condition for stability given by equation (18) can be written as

$$\text{Re}[1 + \Delta_p(e^{j\omega T})] > 0 \quad \text{for all } \omega T. \quad (21)$$

Since the phase of $\Delta_p(e^{j\omega T})$ is unknown, the worst case must be taken in evaluating equation (21), which is when $\Delta_p(e^{j\omega T}) = -B(e^{j\omega T})$. The condition on the unstructured uncertainty bound for stability of the adaptive feedforward controller is thus that

$$B(e^{j\omega T}) < 1 \quad \text{for all } \omega T. \quad (22)$$

3. LINEAR MULTIPLE-INPUT MULTIPLE-OUTPUT PLANT

In a practical control system, the number of plant outputs at the sensors (L) will generally be greater than the number of plant inputs to the actuators (M), so that the matrix of plant responses, $\mathbf{P}(z)$, will not be square. There will also generally be multiple reference, or command, signals (R) so that the matrix of controller responses, $\hat{\mathbf{C}}(z)$, will also not be square. The block diagram of the multichannel feedforward controller can thus be illustrated as in Fig. 3.

The multiple-input multiple-output (MIMO) versions of both the filtered reference and filtered error algorithms will be derived using multichannel generalisations of the summation notation used for the single-input single-output (SISO) case. The block diagrams of these algorithms will then be described in the simple case when there is one command signal and actuator, and two sensors.

The l th error signal at the k th time instant can be written as a generalisation of equation (1),

$$\varepsilon_k(l) = d_k(l) - \sum_{m=1}^M \sum_{j=0}^J p_j(l, m) u_{k-j}(m), \quad (23)$$

where $u_k(m)$ is the m th actuator input and $p_j(l, m)$ is the impulse response of the plant at the l th sensor due to the m th actuator. Similarly the m th actuator input can be written as a generalisation of equation (2) as

$$u_k(m) = \sum_{\kappa=1}^R \sum_{n=0}^N \hat{c}_n(m, \kappa) i_{k-n}(\kappa), \quad (24)$$

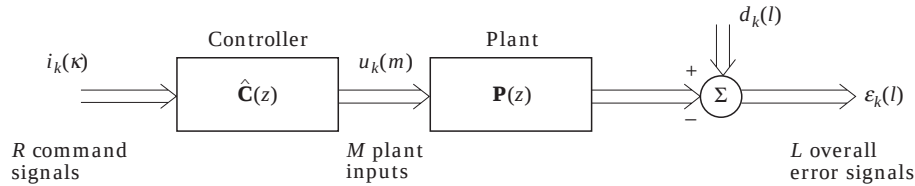


Figure 3. Block diagram of a linear MIMO control system.

where $i_k(\kappa)$ is the κ th command signal and $\hat{c}_n(m, \kappa)$ is the impulse response of the controller at the m th actuator due to the κ th reference signal. The final expression for the overall error is thus rather complicated since it involves summations over reference signals, controller coefficients, secondary actuators and plant coefficients, to give

$$\varepsilon_k(l) = d_k(l) - \sum_{\kappa=1}^R \sum_{n=0}^N \sum_{m=1}^M \sum_{j=0}^J p_j(l, m) \hat{c}_n(m, \kappa) i_{k-n-j}(\kappa). \quad (25)$$

The derivative of the time average sum of squared errors with respect to the n th coefficient of the m, κ th control filter is thus

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \sum_{l=1}^L \varepsilon_k^2(l)}{\partial \hat{c}_n(m, \kappa)} = -\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \sum_{l=1}^L \varepsilon_k(l) \sum_{j=0}^J p_j(l, m) i_{k-m-j}(\kappa) \quad (26)$$

which again can be viewed in two different ways to derive the filtered reference and filtered error algorithms.

3.1. FILTERED REFERENCE ALGORITHM

The l, m, κ th filtered reference signal can be defined as

$$x'_k(l, m, \kappa) = \sum_{j=0}^J \hat{p}_j(l, m) i_{k-j}(\kappa) \quad (27)$$

which is the κ th reference signal filtered by an estimate of the l, m th plant response. The derivative defined in equation (26) is thus the sum of products of L error signals and filtered reference signals. The MIMO filtered reference LMS algorithm can thus be obtained by taking an instantaneous estimate of this average derivative to give [12]

$$\hat{\mathbf{c}}_{k+1} = \hat{\mathbf{c}}_k + \alpha \mathbf{X}'_k \boldsymbol{\varepsilon}_k \quad (28)$$

where $\hat{\mathbf{c}}_k$ is the concatenated vector of N controller coefficients for all $M \times R$ control filters at the k th sample

$$\hat{\mathbf{c}}_k = [\mathbf{c}_k^T(1, 1), \mathbf{c}_k^T(1, 2) \cdots \mathbf{c}_k^T(M, R)]^T, \quad (29)$$

$\hat{\mathbf{c}}_k(m, \kappa) = [\hat{c}_{0,k}(m, \kappa), \hat{c}_{1,k}(m, \kappa) \cdots \hat{c}_{N,k}(m, \kappa)]^T$, $\mathbf{X}'_k$ is a $NMR \times L$ matrix of filtered reference signals whose $nm\kappa, l$ th element is $x'_k(l, m, \kappa)$ and the vector of error signals is

$$\boldsymbol{\varepsilon}_k = [\varepsilon_k(1) \cdots \varepsilon_k(L)]^T. \quad (30)$$

This generalisation of the SISO filtered reference algorithm, equation (6), does not appear to have a simple interpretation in terms of matrix block diagrams. However, it has been used successfully for the multichannel control of sound in aircraft, cars and other applications for over 10 years [12].

3.2. FILTERED ERROR ALGORITHM

Returning to the general expression for the derivative in equation (26), one can again use the dummy time variable defined in equation (8) to re-write this as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \sum_{l=1}^L \varepsilon_k^2(l)}{\partial \hat{c}_n(m, \kappa)} = - \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{j=0}^J \sum_{k'=-K}^K i_{k'}(\kappa) \sum_{l=1}^L \hat{p}_j(l, m) \varepsilon_{k'+n+j}(l). \quad (31)$$

Defining the sum of the reverse-filtered error signals associated with the m th actuator as

$$\delta_k(m) = \sum_{l=1}^L \sum_{j=0}^J \hat{p}_j(l, m) \varepsilon_{k+j}(l) \quad (32)$$

then one can write equation (31), by analogy with equation (12), as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \sum_{l=1}^L \varepsilon_k^2(l)}{\partial \hat{c}_n(m, \kappa)} = - \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \delta_k(m) i_{k-n}(\kappa). \quad (33)$$

The causality and ordering problems can be overcome in the same way as in the SISO case, and the final multichannel filtered error algorithm [13] can now be written in terms of the $M \times RN$ matrix of control filter coefficients as

$$\hat{\mathbf{C}}_{k+1} = \hat{\mathbf{C}}_k + \alpha \delta_{K-J} \mathbf{i}_{K-J}^T \quad (34)$$

where the vector of M delayed filtered error signals is

$$\delta_{K-J} = [\delta_{K-J}(1) \quad \cdots \quad \delta_{K-J}(M)]^T \quad (35)$$

and the vector of RN delayed command signals is

$$\mathbf{i}_{K-J} = [i_{K-J}(1) \quad \cdots \quad i_{K-J-N}(1) \quad i_{K-J}(2) \quad \cdots \quad i_{K-J-N}(R)]^T. \quad (36)$$

The block diagram of the MIMO filtered error controller is shown in Fig. 4.

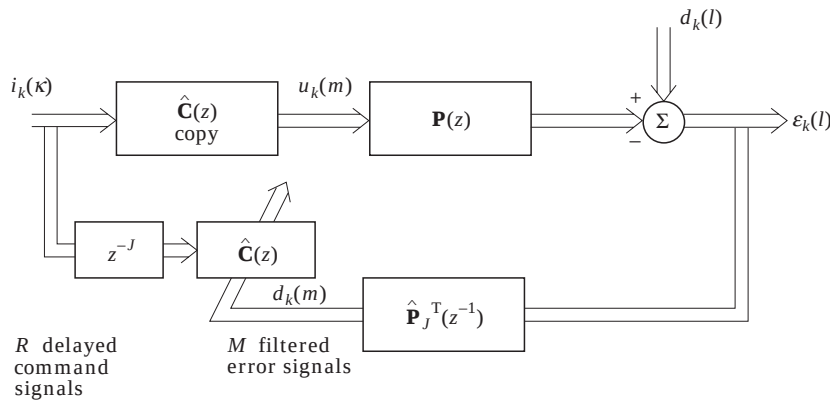


Figure 4. Block diagram of the MIMO filtered error controller adaptation algorithm.

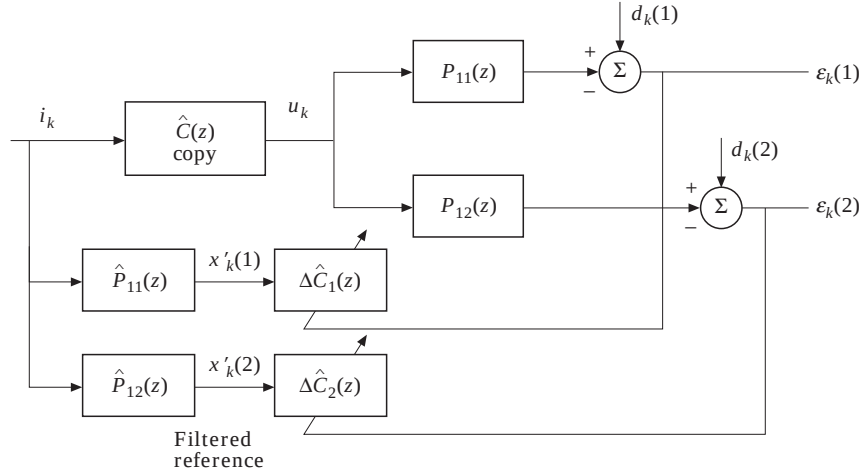


Figure 5. Block diagram for the filtered reference LMS algorithm in a control system with one command signal, one actuator and two sensors, in which both $\Delta \hat{C}_1(z)$ and $\Delta \hat{C}_2(z)$ are used to update $\hat{C}(z)$.

3.3. CONTROL ALGORITHMS FOR A SYSTEM WITH ONLY TWO SENSORS

The main features of the two multichannel adaptation algorithms can be illustrated using the example of a control system with one command signal, one plant input and two plant outputs, in which the object is to minimise the sum of the squared value of the two error signals. The filtered reference adaptation algorithm in this case reduces to

$$\hat{\mathbf{c}}_{k+1} = \hat{\mathbf{c}}_k + \alpha [\varepsilon_k(1) \mathbf{x}'_k(1) + \varepsilon_k(2) \mathbf{x}'_k(2)] \quad (37)$$

where $\mathbf{x}'_k(1)$ and $\mathbf{x}'_k(2)$ are the vectors of filtered reference signals obtained by filtering i_k by estimates of the plant responses from the input to the two outputs, $\hat{P}_{11}(z)$ and $\hat{P}_{12}(z)$. This is illustrated in Fig. 5 as two separate adaptations, to produce the update terms $\Delta \hat{C}_1(z)$ and $\Delta \hat{C}_2(z)$, which are both used to update $\hat{C}(z)$. To extend the already rather ungainly block diagram shown in Fig. 5 to the case of multiple command signals requires multiple copies of the individual paths in the plant model.

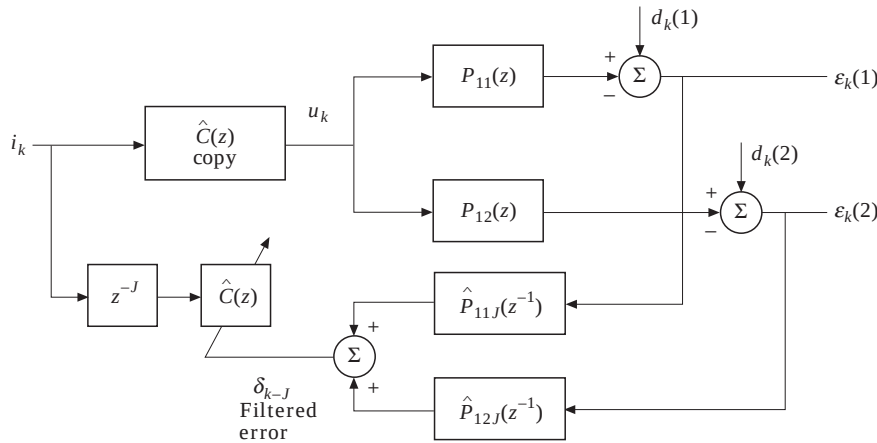


Figure 6. Block diagram for the filtered error LMS algorithm in a control system with one command signal, one actuator and two sensors.

In contrast, the block diagram of Fig. 6, for the filtered error algorithm in this case, is fairly neat and can be readily extended to the case of multiple reference signals. This is because the adaptation only depends on one product, as in equation (33), although one signal in the product is the sum of two reverse-filtered error signals, which in this case is given by

$$\delta_{k-J} = \sum_{j=0}^J p_{J-j}(1)e_{k-j}(1) + \sum_{j=0}^J p_{J-j}(2)e_{k-j}(2). \quad (38)$$

In general the filtered reference algorithm thus requires R sets of LM filtering operations whereas the filtered error algorithm only requires a single set of LM filtering operations. In control systems with several command signals ($R > 1$), the filtered error algorithm thus has a considerably smaller processing requirement.

4. NON-LINEAR PLANT AND/OR CONTROLLER

This section considers the steepest-descent adaptation of a coefficient, or weight w , embedded in a non-linear system which would typically be the combination of a non-linear plant and a non-linear controller. Such an arrangement is illustrated in Fig. 7 in which the non-linear system is assumed to be entirely feedforward and driven by a set of command signals $\mathbf{i}_k$. If one assumes that the weight is situated at one point in a particular layer of the feedforward controller, the overall error signal will be a function of the output signals from this layer, e.g. b_k , and the input signals to this layer, e.g. a_k , will only depend on the command signals. a_k is weighted by w to produce the output signal

$$b_k = wa_k, \quad (39)$$

which influences each of the output signals from the network, $\mathbf{y}_k$, which are mixed with the desired signals, $\mathbf{d}_k$, to form the vector of L error signals,

$$\boldsymbol{\varepsilon}_k = \mathbf{d}_k - \mathbf{y}_k = [\varepsilon_k(1) \quad \cdots \quad \varepsilon_k(L)]^T. \quad (40)$$

The object is to minimise the sum of the squared values of these error signals using the gradient

$$\frac{\partial \boldsymbol{\varepsilon}_k^T \boldsymbol{\varepsilon}_k}{\partial w} = -2\boldsymbol{\varepsilon}_k^T \frac{\partial \mathbf{y}_k}{\partial w}. \quad (41)$$

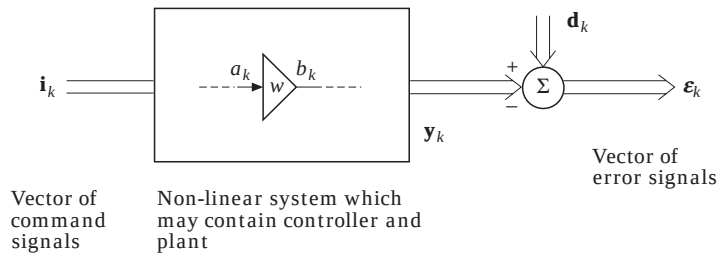


Figure 7. Block diagram of a non-linear MIMO system.

Now it is assumed that $\mathbf{y}_k$ only depends on J past values of b_k , where again J could be arbitrarily large, so that

$$\frac{\partial \mathbf{y}_k}{\partial w} = \sum_{j=0}^J \frac{\partial \mathbf{y}_k}{\partial b_{k-j}} \frac{\partial b_{k-j}}{\partial w} = \sum_{j=0}^J \frac{\partial \mathbf{y}_k}{\partial b_{k-j}} a_{k-j}. \quad (42)$$

Alternatively, if the network is recurrent, then this derivative can be calculated recursively [14].

4.1. FILTERED REFERENCE ALGORITHM

The terms $\partial \mathbf{y}_k / \partial b_{k-j}$ in equation (42) correspond to the differential change in each of the current system outputs due to a perturbation in the signal b_k j samples ago. Thus, they can be interpreted as the small-signal impulse responses of the multichannel non-linear system which is linearised about the current operating point. Because this impulse response changes with time due to changes in the operating point, defined by the other weights and signals in the controller to the right of w in Fig. 7, then a non-linear problem has effectively been traded for a time-varying one. These terms in equation (42) are written as

$$\frac{\partial \mathbf{y}_k}{\partial b_{k-j}} = \mathbf{p}_j \quad (43)$$

which is a slowly varying function of time if the non-linear system does not have discontinuous behaviour. Thus equation (42) can be written as

$$\frac{\partial \mathbf{y}_k}{\partial w} = \sum_{j=0}^J \mathbf{p}_j a_{k-j} \quad (44)$$

which defines a vector of filtered reference signals, which is equal to the input signal to the weight w filtered by the linearised response from the output of the weight to the vector of system outputs, and can thus be written as $\mathbf{x}'_k$.

A filtered reference adaptation algorithm for w using the instantaneous gradient defined in equation (41) can thus be written as

$$w_{k+1} = w_k + \alpha \mathbf{e}_k^T \mathbf{x}'_k. \quad (45)$$

Adaptation algorithms of this type for the weights in neural networks have been interpreted by Marcos *et al.* [15] and Beaufays and Wan [14] as a form of the real-time backpropagation (RTBP) algorithm [16].

4.2. FILTERED ERROR ALGORITHM

Using equation (44), the time-averaged version of the derivative of equation (41) can also be written as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \mathbf{e}_k^T \mathbf{e}_k}{\partial w} = - \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=-K}^K \sum_{l=1}^L \mathbf{e}_k(l) \sum_{j=0}^J p_j(l) a_{k-j}. \quad (46)$$

Again using the dummy time variable defined by equation (8), equation (46) can be written as

$$\lim_{K \rightarrow \infty} \frac{1}{2K} \sum_{k=-K}^K \frac{\partial \mathbf{e}_k^T \mathbf{e}_k}{\partial w} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k'=-K}^K a_{k'} \sum_{l=0}^L \sum_{j=0}^j p_j(l) \mathbf{e}_{k'+j}(l). \quad (47)$$

The sum of the reverse filtered error signal can be written as

$$\delta_k = \sum_{l=0}^L \sum_{j=0}^J p_j(l) \varepsilon_{k+j}(l) \quad (48)$$

which is consistent with the definition used in [8], for example, which is of the form

$$\delta_k = \frac{\partial \sum_{l=1}^L \varepsilon_k^2(l)}{\partial b_k}. \quad (49)$$

The instantaneous version of equation (47), after reordering and delaying, as above, can thus be used to derive an instantaneous steepest descent algorithm for the adaptation of w in this case, which has the form

$$w_{k+1} = w_k + \alpha \delta_{k-J} a_{k-J}. \quad (50)$$

Adaptive algorithms of this type for neural networks have been interpreted by Marcos *et al.* [15] and Beaufays and Wan [14] as a form of backpropagation through time (BPTT) [17]. This algorithm is usually used to perform a block adaptation based on the aggregate update term, in terminal control problems for example, rather than in the instantaneous form of equation (50). Beaufays and Wan [14] also showed that the reverse filtering operation of equation (48) can be interpreted as the flow of a signal through the transposed or reciprocal of the original non-linear network.

5. CONCLUSION

A consistent formulation has been presented for filtered reference and filtered error forms of the LMS algorithm used in feedforward control. Both algorithms minimise the mean-square value of the same, output, error. The derivation of the algorithms is clearest for the SISO case, but can be generalised to cover both MIMO plants with multiple reference signals, and the case of a non-linear controller and/or plant.

The simple block diagram for the SISO filtered error algorithm can be generalised using matrix operations to the MIMO case. The block diagram of the MIMO filtered reference algorithm is not so elegant, but the algorithm can be readily implemented by using a formulation using summations, and has been found to work well in practice. For non-linear systems the filtered reference and filtered error LMS algorithms lead to two different forms of the backpropagation algorithm widely used to adapt neural networks.

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